

A Robust and Highly Efficient Approach for Isolation of Mesenchymal Stem Cells From Wharton's Jelly for Tissue Repair.

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Mesenchymal stem cells derived from umbilical cord Wharton's Jelly (WJ-MSCs) are emerging as promising therapeutics for a variety of diseases due to their ability of regeneration and immunomodulation, and their non-tumorigenic and non-immunogenic properties. Although multiple protocols have been developed for WJ-MSC isolation, insufficient cell numbers, heterogeneous cell population, and variations in procedures between different laboratories impede further clinical applications. Here, we compared six widely used WJ-MSC isolation methods regarding cell morphology, yield, purity, proliferation rate, and differentiation potential. Based on these analyses, we identified that the inefficiency of the extracellular matrix digestion results in low cell yield. Thus, we developed a new method called "Mince-Soak-Digest (MSD)" to isolate MSCs from WJ by incorporating a soaking step to facilitate the digestion of the extracellular matrix and release of the cells. Our newly developed method generates significantly higher cell yield (4- to 10-fold higher) than six widely used methods that we tested with high purity and consistency. Importantly, by transplantation of WJ-MSCs to the rat uterus, we repair the endometrial injury and restore the fertility of the rats. In conclusion, our results provide a robust and highly efficient approach for the isolation of WJ-MSCs to restore injured tissue. The higher efficiency of MSD assures the abundance of WJ-MSCs for clinical applications. Furthermore, the reliability of MSD contributes to the standardization of WJ-MSC isolation, which eliminates the discrepancies due to isolation procedures, thus facilitating the evaluation of the efficacy of WJ-MSCs across various human clinical applications.

Human umbilical cord Wharton's jelly-derived mesenchymal stem cell transplantation could improve diabetic intracavernosal pressure.

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Mesenchymal stem cells (MSCs) secrete various cytokines with angiogenic and neuroprotective effects. This study aimed to assess the effects of human umbilical cord Wharton's jelly-derived MSCs (hWJ-MSCs) on diabetes-related intracavernosal pressure (ICP) impairment in rats. hWJ-MSCs were isolated from human umbilical cord Wharton's jelly and transplanted into the corpus cavernosum of streptozotocin (STZ)-induced diabetic rats by unilateral injection. The erectile function was evaluated at 4 weeks, as well as the expression levels of vascular endothelial growth factor (VEGF), basic fibroblast growth factor (bFGF), endothelial nitric oxide synthase (eNOS), and insulin-like growth factor 1 (IGF1). STZ-induced diabetic rats showed impaired ICP, which was significantly improved by hWJ-MSC treatment. VEGF, eNOS, IGF1, and bFGF expression levels were higher in hWJ-MSC injection sites than those in control ones in STZ-induced diabetic rats. These results suggest that hWJ-MSC transplantation might improve diabetic erectile dysfunction through increased production of paracrine growth factors, highlighting a novel potential therapeutic option for erectile dysfunction.

Wharton's jelly derived mesenchymal stromal cells: Biological properties, induction of neuronal phenotype and current applications in neurodegeneration research.

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Multipotent mesenchymal stromal cells, also known as mesenchymal stem cells (MSC), can be isolated from bone marrow or other tissues, including fat, muscle and umbilical cord. It has been shown that MSC behave in vitro as stem cells: they self-renew and are able to differentiate into mature cells typical of several mesenchymal tissues. Moreover, the differentiation toward non-mesenchymal cell lineages (e.g. neurons) has been reported as well. The clinical relevance of these cells is mainly related to their ability to spontaneously migrate to the site of inflammation/damage, to their safety profile thanks to their low immunogenicity and to their immunomodulation capacities. To date, MSCs isolated from the post-natal bone marrow have represented the most extensively studied population of adult MSCs, in view of their possible use in various therapeutical applications. However, the bone marrow-derived MSCs exhibit a series of limitations, mainly related to their problematic isolation, culturing and use. In recent years, umbilical cord (UC) matrix (i.e. Wharton's jelly, WJ) stromal cells have therefore emerged as a more suitable alternative source of MSCs, thanks to their primitive nature and the easy isolation without relevant ethical concerns. This review seeks to provide an overview of the main biological properties of WJ-derived MSCs. Moreover, the potential application of these cells for the treatment of some known dysfunctions in the central and peripheral nervous system will also be discussed.

Wharton's Jelly stem cells: future clinical applications.

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This review focuses on the therapeutic potential of stem cells harvested from the Wharton's Jelly of the human umbilical cord. Recently, investigators have found that a potent stem cell population exists within the Wharton's Jelly. In this review, the authors define a new subset of stem cells, termed perinatal stem cells, and compare them to other sources of stem cells. Furthermore, cryopreservation of Wharton's Jelly stem cells is described for potential use in future cell based therapies and/or regenerative medicine applications. Current evidence of the application of mesenchymal stem cells from various sources in both pre-clinical and clinical trials is reviewed in the context of potential indications of use for Wharton's Jelly derived mesenchymal stem cells.

Human umbilical cord Wharton's Jelly-derived mesenchymal stem cells differentiation into nerve-like cells.

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The two most basic properties of mesenchymal stem cells (MSCs) are the capacities to self-renew indefinitely and differentiate into multiple cells and tissue types. The cells from human umbilical cord Wharton's Jelly have properties of MSCs and represent a rich source of primitive cells. This study was conducted to explore the possibility of inducing human umbilical cord Wharton's Jelly-derived MSCs to differentiate into nerve-like cells. MSCs were cultured from the Wharton's Jelly taken from human umbilical cord of babies delivered after full-term normal labor. Salvia miltiorrhiza and beta-mercaptoethanol were used to induce the human umbilical cord-derived MSCs to differentiate. The expression of neural protein markers was shown by immunocytochemistry. The induction process was monitored by phase contrast microscopy, electron microscopy (EM), and laser scanning confocal microscopy (LSCM). The pleiotrophin and nestin genes were measured by reverse

transcription-polymerase chain reaction (RT-PCR). MSCs in the Wharton's Jelly were easily attainable and could be maintained and expanded in culture. They were positive for markers of MSCs, but negative for markers of hematopoietic cells and graft-versus-host disease (GVHD)-related cells. Treatment with *Salvia miltiorrhiza* caused Wharton's Jelly cells to undergo profound morphological changes. The induced MSCs developed rounded cell bodies with multiple neurite-like extensions. Eventually they developed processes that formed networks reminiscent of primary cultures of neurons. *Salvia miltiorrhiza* and beta-mercaptoethanol also induced MSCs to express nestin, beta-tubulinIII, neurofilament (NF) and glial fibrillary acidic protein (GFAP). It was confirmed by RT-PCR that MSCs could express pleiotrophin both before and after induction by *Salvia miltiorrhiza*. The expression was markedly enhanced after induction and the nestin gene was also expressed. MSCs could be isolated from human umbilical cord Wharton's Jelly. They were capable of differentiating into nerve-like cells using *Salvia miltiorrhiza* or beta-mercaptoethanol. The induced MSCs not only underwent morphologic changes, but also expressed the neuron-related genes and neuronal cell markers. They may represent an alternative source of stem cells for central nervous system cell transplantation.

Easy and Rapid Methods for Human Umbilical Cord Blood-Derived Mesenchymal Stem Cells and Human Umbilical Wharton's Jelly-Derived Mesenchymal Stem Cells.

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These protocols describe modified methods that use Ficoll-Paque density gradient for umbilical cord blood-derived mesenchymal stem cells and explant method for Wharton's jelly-derived mesenchymal stem cells. The Ficoll-Paque density gradient method allows to obtain mesenchymal stem cells while eliminating monocytic cells. In this method, precoating the cell culture flasks with fetal bovine serum helps remove the monocytic cells and instruct more pure mesenchymal stem cells. On the other hand, the explant method for Wharton's jelly-derived mesenchymal stem cell is user-friendly and cost-effective than enzymatic methods. In this chapter, we provide a collection of protocols to obtain mesenchymal stem cells from human umbilical cord blood and Wharton's jelly.

Isolation and Characterization of Feline Wharton's Jelly-Derived Mesenchymal Stem Cells.

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Wharton's jelly is a well-known mesenchymal stem cell source in many species, including humans. However, there have been no reports confirming the presence of mesenchymal stem cells in Wharton's jelly in cats. The purpose of this study was to isolate mesenchymal stem cells (MSCs) from the Wharton's jelly of cats and to characterize stem cells. In this study, feline Wharton's jelly-derived mesenchymal stem cells (fWJ-MSCs) were isolated and successfully cultured. fWJ-MSCs were maintained and the proliferative potential was measured by cumulative population doubling level (CPDL) test, scratch test, and colony forming unit (CFU) test. Stem cell marker, karyotyping and immunophenotyping analysis by flow cytometry showed that fWJ-MSCs possessed characteristic mesenchymal stem cell markers. To confirm the differentiation potential, we performed osteogenic, adipogenic and chondrogenic induction under each differentiation condition. fWJ-MSCs has the ability to differentiate into multiple lineages, including osteogenic, adipogenic and chondrogenic differentiation. This study shows that Wharton's jelly

of cat can be a good source of mesenchymal stem cells. In addition, fWJ-MSCs may be useful for stem cell-based therapeutic applications in feline medicine.

Ectodermal Differentiation of Wharton's Jelly Mesenchymal Stem Cells for Tissue Engineering and Regenerative Medicine Applications.

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Mesenchymal stem cells (MSCs) from Wharton's jelly (WJ) of the human umbilical cord are perinatal stem cells that have self-renewal ability, extended proliferation potential, immunosuppressive properties, and are accordingly excellent candidates for tissue engineering. These MSCs are unique, easily accessible, and a noncontroversial cell source of regeneration in medicine. Wharton's jelly mesenchymal stem cells (WJMSCs) are multipotent and capable of multilineage differentiation into cells like adipocytes, bone, cartilage, and skeletal muscle upon exposure to appropriate conditions. The ectoderm is one of the three primary germ layers found in the very early embryo that differentiates into the epidermis, nervous system (spine, peripheral nerves, brain), and exocrine glands (mammary, sweat, salivary, and lacrimal glands). Accumulating evidence shows that MSCs obtained from WJ have an ectodermal differentiation potential. The current review examines this differentiation potential of WJMSC into the hair follicle, skin, neurons, and sweat glands along with discussing the potential utilization of such differentiation in regenerative medicine.

Human Wharton's Jelly-Cellular Specificity, Stemness Potency, Animal Models, and Current Application in Human Clinical Trials.

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Stem cell therapies offer a great promise for regenerative and reconstructive medicine, due to their self-renewal and differentiation capacity. Although embryonic stem cells are pluripotent, their utilization involves embryo destruction and is ethically controversial. Therefore, adult tissues that have emerged as an alternative source of stem cells and perinatal tissues, such as the umbilical cord, appear to be particularly attractive. Wharton's jelly, a gelatinous connective tissue contained in the umbilical cord, is abundant in mesenchymal stem cells (MSCs) that express CD105, CD73, CD90, Oct-4, Sox-2, and Nanog among others, and have the ability to differentiate into osteogenic, adipogenic, chondrogenic, and other lineages. Moreover, Wharton's jelly-derived MSCs (WJ-MSCs) do not express MHC-II and exhibit immunomodulatory properties, which makes them a good alternative for allogeneic and xenogeneic transplantations in cellular therapies. Therefore, umbilical cord, especially Wharton's jelly, is a promising source of mesenchymal stem cells.

Wharton's jelly-derived mesenchymal stem cells: phenotypic characterization and optimizing their therapeutic potential for clinical applications.

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Wharton's jelly (WJ) is a gelatinous tissue within the umbilical cord that contains myofibroblast-like stromal cells. A unique cell population of WJ that has been suggested as displaying the stemness phenotype is the mesenchymal stromal cells (MSCs). Because MSCs' stemness and immune properties appear to be more robustly expressed and functional which are more comparable with fetal than adult-derived MSCs, MSCs harvested from the "young" WJ are considered much more proliferative, immunosuppressive, and even therapeutically active stem cells than those isolated from older, adult tissue sources such as the bone marrow or adipose. The present review discusses the phenotypic characteristics, therapeutic applications, and optimization of experimental protocols for WJ-derived stem cells. MSCs derived from WJ display promising transplantable features, including ease of sourcing, in vitro expandability, differentiation abilities, immune-evasion and immune-regulation capacities. Accumulating evidence demonstrates that WJ-derived stem cells possess many potential advantages as transplantable cells for treatment of various diseases (e.g., cancer, chronic liver disease, cardiovascular diseases, nerve, cartilage and tendon injury). Additional studies are warranted to translate the use of WJ-derived stem cells for clinical applications.
